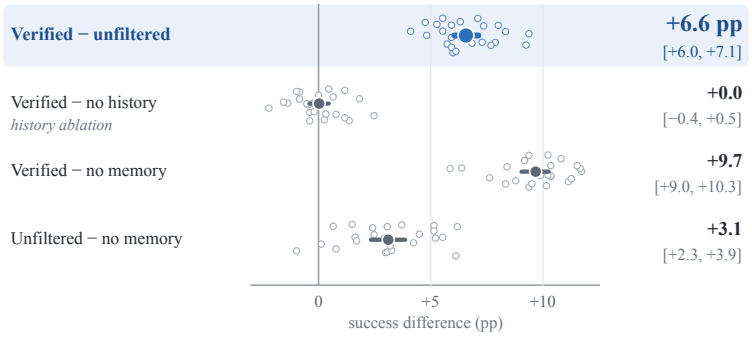
(a) Paired success changes

percentage points · ○ = 24 seed blocks · bar = paired 95% CI



(b) Gain by task and budget

SYNTHETIC DATA

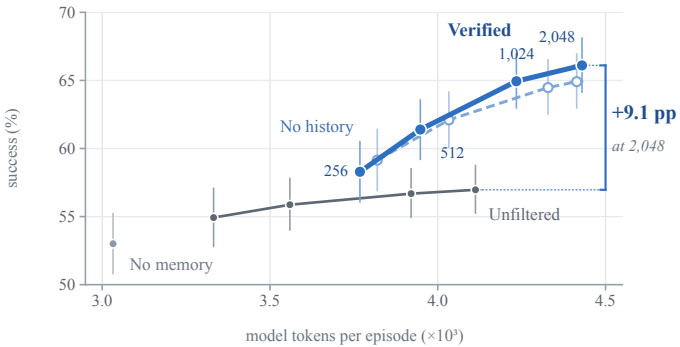
verified − unfiltered, pp · ○ = CI includes 0

	memory budget (tokens) 256	512	1,024	2,048
Fact recall	+0.3 ○	+0.5 ○	+1.3	+1.9
Code repair	+4.1	+6.9	+9.9	**+11.1**
Tool routing	+5.4	+8.6	**+11.6**	**+12.5**
Long horizon	+3.6	+6.0	**+10.2**	**+11.1**

range **+0.3** (fact recall, 256) to **+12.5** (tool routing, 2,048)

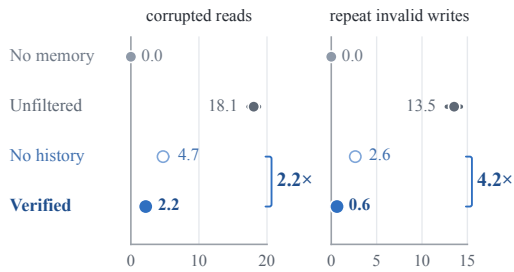
(c) Success versus token cost

mean over task families · budgets 256–2,048 tokens · 95% CI on both axes



(d) Rejection-history diagnostic

events per 100 episodes · history uses 25% of capacity



fewer errors with history, yet success **+0.0 pp** in (a)